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Hydrogen Evolution Reaction | Hot Paper

A Hydrogen Evolution Catalyst [Co_2O_2] Metallacycle Enables Regioselective Allene C(sp^2)-H Functionalization

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In memory of Professor Xiyan Lu.

Abstract: Selective functionalization of allenic C(sp^2)-H is an ideal approach to upgrading simple allenes to synthetically useful allenes, albeit suffering from challenges associated with inert reactivity and inferior selectivity. Inspired by energy chemistry, a catalytic hydrogen evolution reaction (HER) strategy was leveraged to selectively activate weakly acidic allene C(sp^2)-H bonds in a reductive mode. An array of [Co_2O_2] metallacycle complexes were readily devised starting from amino acids, and they were demonstrated as robust HER catalysts, which would selectively break allenic C(sp^2)-H bonds to release hydrogen. With the newly developed HER catalyst, regioselective electrochemical functionalization of allenic C(sp^2)-H with alcoholic α C(sp^3)-H was unprecedentedly achieved. This strategy features excellent regioselectivity, unconventional chemoselectivity, good functional-group tolerance (62 examples), and mild conditions. Mechanism experiments revealed a reactive hydroxy-coordinated cobalt(II) species in the reaction. Density functional theory (DFT) calculations were also conducted to rationalize the regioselectivity observed in the reaction.

Introduction

Allene is one of the most versatile building blocks and widely distributed in natural products and pharmaceuticals.[1–5] The functionalization of allenes is an attractive and direct approach to allene derivatization (Figure 1A). Although the hydrofunctionalization of allenes pioneered by Krische[6–11] has been well-documented, the direct functionalization of allenic C(sp^2)-H bonds still suffers from some challenging issues in the transformation associated with the inert reactivity and inferior selectivity. To circumvent these issues, activated allenes bearing electron-withdrawing groups[12–17] or 1,1-disubstituted allenes[18–19] are commonly required in the previous reports. Recently, Wang[20] elegantly developed a strategy to enhance the acidity of simple allenes using a cationic cyclopentadienyliron catalyst (Figure 1B). Almost at the same time, Carreira[21] group unveiled a palladium-catalyzed allene C–H functionalization with the assistance of a picolinamide directing group (Figure 1C). Very recently, remarkable progress in the selective cyanation of allenic C(sp^2)-H has been independently achieved by Liu[22] and Wu[23] through a copper-catalyzed radical relay strategy (Figure 1D). Despite the prominent achievements, these approaches involve sophisticated catalysts, stoichiometric oxidants, or high temperatures; a general and mild approach is still in high demand. As a part of our continuous interests in exploring the potential of electrochemistry in organic synthesis,[24–29] we envisaged that synthetic electrochemistry[30–40] might offer an alternative solution for this transformation.

Hydrogen evolution reaction (HER)[41–46] is one of the most fundamental transformations that convert energy from electricity into fuels. The development of HER catalysts featuring high efficiency and availability would significantly revolutionize energy storage and delivery systems, and tremendous progress has been achieved. Moreover, drawing inspiration from catalytic HER would bring changes to the challenges in synthetic chemistry. Recently, Baran[47] and Lin[48] independently reported electrochemical protocols to intercept cobalt hydride species in cobalt-catalyzed HER. Nevertheless, using HER catalysts to leverage organic transformations directly still received far less attention. From the perspective of organic synthesis, HER provides a reductive activation mode for allenic C(sp^2)-H by shifting its dissociation equilibrium; allenyl carbanions[49] would be generated in high regioselective control, thus facilitating their selective functionalization (Figure 1E). Discovering a simple and efficient HER catalyst is critical for the success of the electrochemical transformations.

In previous reports, DuBois and co-workers[50] demonstrated that amine base sites as proton relays in HER catalyst could significantly improve its catalytic performance (Figure 1F). Inspired by the work, [Co_2O_2] metallacycle complexes of amino acids containing multiple base

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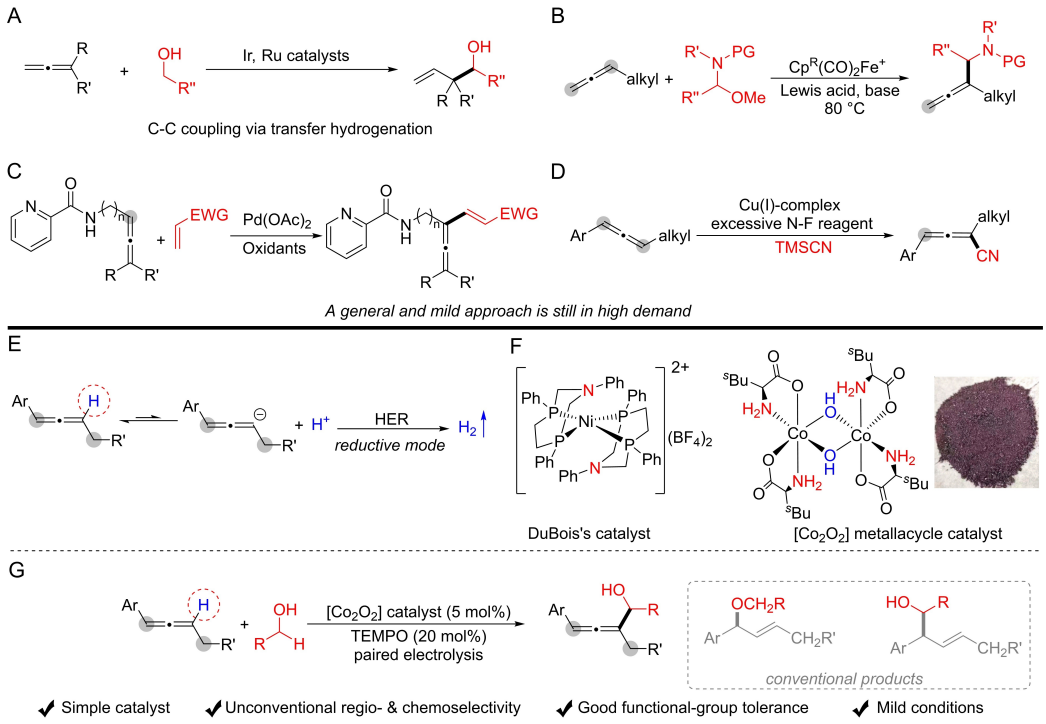


Figure 1. Approaches for the selective functionalization of allenes. (A) Krische's work. (B) Wang's work. (C) Carreira's work. (D) Liu and Wu's work. (E) A reductive activation mode for allene C(sp^2)-H enabled by HER. (F) HER catalysts bearing alkaline sites. (G) This work.

sites (OH, amine) were postulated as efficient HER catalysts (Figure 1F). Although the μ-OH bridged binuclear cobalt(III) complexes have been investigated in coordination chemistry,[51–52] their HER catalytic performance remains elusive. In this work, these readily available and scalable complexes showed robust HER catalytic activity and provided a mild avenue for allenyl carbanions. By virtue of the novel catalyst and a cocatalyst 2,2,6,6-tetramethylpiperidoxyl (TEMPO),[53] electrochemical functionalization of allenic C(sp^2)-H with alcohols was readily achieved in high regioselectivity and unconventional chemoselectivity (Figure 1G), which previously only afford O–C coupling[54–55] and hydrofunctionalization[6–11] products. This electrochemical protocol exhibits some impressive advantages: a) simple catalyst; b) unconventional regio- and chemoselectivity; c) good functional-group tolerance (62 examples); d) mild conditions.

Results and Discussion

Catalyst synthesis and electrocatalytic performance investigation

According to the previous reports,[51–52] we prepared a broad range of [Co_2O_2] metallacycle complexes in a facile procedure involving deprotonation, coordination, and in situ air oxidation (Figure 2A). Further centrifugation and evaporation of the reaction mixture gave purified products by removing KCl precipitate. The [Co_2O_2] metallacycle structure was unambiguously identified by the single crystal diffraction of **cat 2**.[56] As shown in Figure 2B, the [Co_2O_2] metallacycle core is composed of μ-OH bridged binuclear cobalt(III) centers, which endows the catalyst with alkalinity and promote the proton dissociation of weakly acidic substrates. The catalyst structure was further characterized by nuclear magnetic resonance (NMR) spectra and liquid phase mass spectrum (LC-MS) (Figure S2–S17).

Reduction potentials (Co^{II}/Co^{I}) of catalysts have been listed below (Figure 2C), as they directly determine the catalytic HER[41–46] performance. Two close peaks around −2.0 V were detected for **cat 1** (vs. Fc/Fc^+), and they were tentatively assigned to the cobalt(II) complexes **A** and **B**, which were in situ formed from the reduction of **cat 1** monomer. The LC-MS (Figure S30–S31) of the cathodically reduced solution confirms the existence of **A** and **B**. To clarify the species **A** and **B**, we synthesized them under a nitrogen atmosphere, and they were characterized by LC-MS (Figure S32–S33). We compared the cyclic voltammogram of **cat 1** with that of species **A** and **B** (Figure 2D). The well accordance of the cathodic peaks reveals that the peak at −1.75 V and −2.09 V is generated from **A** and **B**, respectively.

With the well-defined complexes, a titration experiment was next investigated to understand their catalytic performance in the HER of allene **1a** (Figure 2E). It showed that titrating **cat 1** with allene **1a** led to a significant increase of both HER peak of **1a** (−2.34 V) and the reduction peaks (Co^{II}/Co^{I}) arising from **A** (−1.75 V). Notably, species **B** was found to be the inactive species in the catalytic HER

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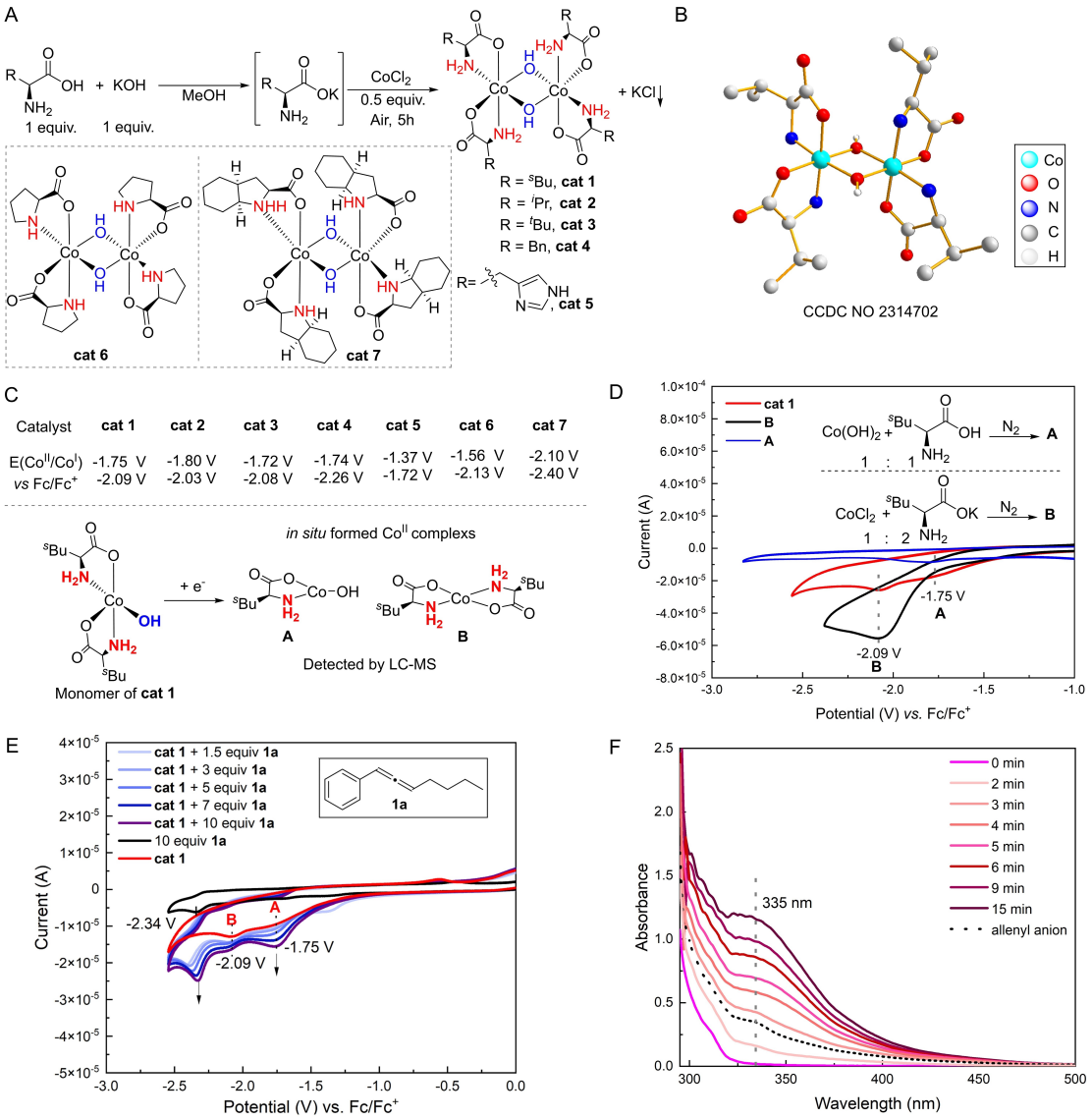


Figure 2. Catalyst synthesis and electrocatalytic performance investigation. (A) Synthetic route to $[Co_2O_2]$ metallacycle catalysts. (B) Single crystal structure of **cat 2**. (C) Redox properties of catalysts. (D) Comparison of cyclic voltammograms. (E) Titration experiment. (F) In situ UV-vis spectrum of electrolyzed solution of **1a**.

with a negligible increase of cathodic peak (−2.09 V). This result supports the important role of the hydroxy group in the catalyst. The HER process was also confirmed by the observation of hydrogen (Figure S38). We also monitored the electrolysis of **1a** in the presence of **cat 1** using in situ ultraviolet-visible (UV-vis) spectroelectrochemistry (Figure 2F). A peak at 335 nm was detected, and its intensity increased with electrolysis time. This species was identified as the allenyl carbanions by making a comparison with the species generated via ^{n}BuLi. Taken together, these observations indicate that the $[Co_2O_2]$ metallacycle catalyst could significantly facilitate the HER of allene **1a**, giving allenyl carbanions via the reactive species **A**.

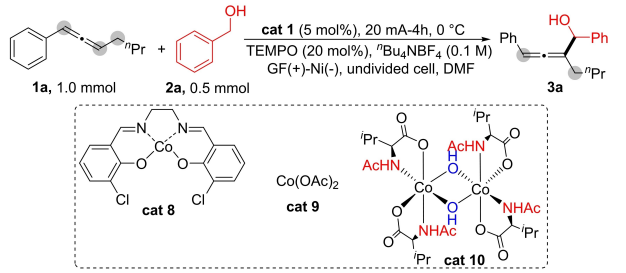
Reaction optimization and substrate scope examination

With the promising HER catalytic performance of the $[Co_2O_2]$ metallacycle complex, we next investigated the electrochemical functionalization of allene **1a**, which contains two similar allenic C(sp^2)-H bonds, with benzyl alcohol **2a** (Table 1). To promote the anodic oxidation of benzyl alcohol, 2,2,6,6-tetramethylpiperidoxyl (TEMPO) was introduced as a cocatalyst in the paired electrolysis system. To our delight, a regioselective product **3a** was exclusively observed in 80 % yield by using **cat 1** as HER catalyst, DMF as solvent, ^{n}Bu$_4$NBF$_4$ as electrolyte, graphite felt and nickel plate as anode and cathode, respectively (Entry 1). Further varying the structure of $[Co_2O_2]$ catalysts failed to give better yields (Entries 2–7). A suitable steric

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Table 1: The reaction conditions optimization of electrochemical functionalization of allene **1a** with benzyl alcohol **2a**.[a]



Entry	Deviation from standard conditions	Yield (%)[b]
1	none	80
2	**cat 2** instead of **cat 1**	78
3	**cat 3** instead of **cat 1**	68
4	**cat 4** instead of **cat 1**	25
5	**cat 5** instead of **cat 1**	27
6	**cat 6** instead of **cat 1**	71
7	**cat 7** instead of **cat 1**	53
8	**cat 8** instead of **cat 1**	30
9	**cat 9** instead of **cat 1**	trace
10	**cat 10** instead of **cat 1**	39
11	removal of **cat 1** or TEMPO	trace

[a] Reaction conditions: **1a** (1.0 mmol), **2a** (0.5 mmol), **cat 1** (0.025 mmol), TEMPO (0.1 mmol), DMF (10 mL), $^{n}Bu_4NBF_4$ (1.0 mmol), graphite felt anode, nickel plate cathode, undivided cell, 0 °C, 20 mA, 4 h (5.97 F/mol). [b] Isolated yield.

hindrance might be critical for the catalyst's performance as the structurally similar L-Valine catalyst (**cat 2**) gave a comparable yield (Entry 2). Replacing [Co_2O_2] metallacycle catalysts with a conventional HER catalyst (**cat 8**) led to significantly diminished yield (30 %, Entry 8), suggesting the superiority of the newly developed catalysts. To demonstrate the crucial role of amine sites in the catalyst, **cat 9** and **cat 10** were investigated in the reaction, and product **3a** was delivered in trace and 39 % yield, respectively (Entries 9–10). This experimental observation supports the necessity of amine sites in the catalyst. The control experiment removed either HER catalyst (**cat 1**) or TEMPO shut down the reaction directly (Entry 11). Under the optimal conditions, we quantified the byproduct hydrogen released from the reaction with a GC analysis (Figure S39), and around 0.51 mmol hydrogen was determined to support the acceptorless dehydrogenation process.

Having identified the optimal conditions, we turned our attention to examining substrate generality (Scheme 1). A broad range of alcohol substrates were employed in the electrochemical protocol. Initially, we investigated the electronic effect by using *para*-substituted benzyl alcohols. It was found that electron-donating groups (**3a**–**3h**) were favored in the reaction compared with electron-withdrawing ones (**3i**–**3o**), presumably due to their lower oxidation potential. We replaced **cat 1** with **cat 2** to overcome the inertness of the electron-deficient substrates. Subsequently, the superiority of our protocol was further demonstrated with some challenging but useful functional groups such as thioether (**3p**), vinyl (**3q**), amine (**3r**), iodine (**3s**), carboxylate (**3t**) and boric ester (**3u**); corresponding products were readily observed albeit with lower yields. The hydrolysis of ester groups induced by the alkalinity of allenyl carbanions gives rise to deteriorated yields (**3t**–**3u**). Other substitution patterns (**3v**–**3z**), fused rings (**3aa**–**3ad**), and heterocycles (**3ae**–**3af**) were tested with the optimal conditions, and largely maintained yields were observed. Furthermore, aliphatic alcohols (**3ag**–**3ak**) were also found to be suitable substrates to deliver desired products with synthetically useful yields (30–73 %). Specifically, cyclopropylmethanol affords the desired product **3ag** in 73 % without detecting radical-initiated ring expansion product, suggesting a possible ionic pathway. This electrochemical approach enables a platform for the late-stage functionalization of the pharmaceutical Adapalene-derived alcohol (**3al**). Remarkably, our approach could readily discriminate two alcoholic α C–H bonds to give the site-selective products **3am**–**3an**. Nevertheless, secondary alcohols failed to afford the desired products.

To further illustrate the generality of the electrochemical protocol, various allenes were synthesized and tested in the reaction. As shown in Scheme 2, changing aliphatic chain R to other groups (**3ao**–**3as**) marginally affected the reaction yield. Notably, allene bearing a cyclopropyl group was employed to probe the reaction mechanism, and the desired product **3ar** was afforded in 75 % yield. Investigation on the allene substitution revealed that 1,1-disubstituted allene (**3at**) and trisubstituted allene (**3au**) were amenable to afford the desired products, albeit with lower yields. The electronic effect (**3av**–**3bd**) and position of substituents (**3be**–**3bf**) were next explored. In general, this electrochemical approach showed good tolerance for these variations, and corresponding products were accessed in satisfactory yields. We also tested the site selectivity of the reaction using an allyloxy substituted allene, and the allenic C(sp^2)-H functionalization product **3bg** was exclusively accessed. Substrates containing naphthyl (**3bh**) or thienyl (**3bi**–**3bj**) groups proceeded smoothly in the reaction to give desired allenol products. Alkyl disubstituted allenes and tertiary butyl allene proved unsuccessful substrates due to their weak acidity or steric hindrance.

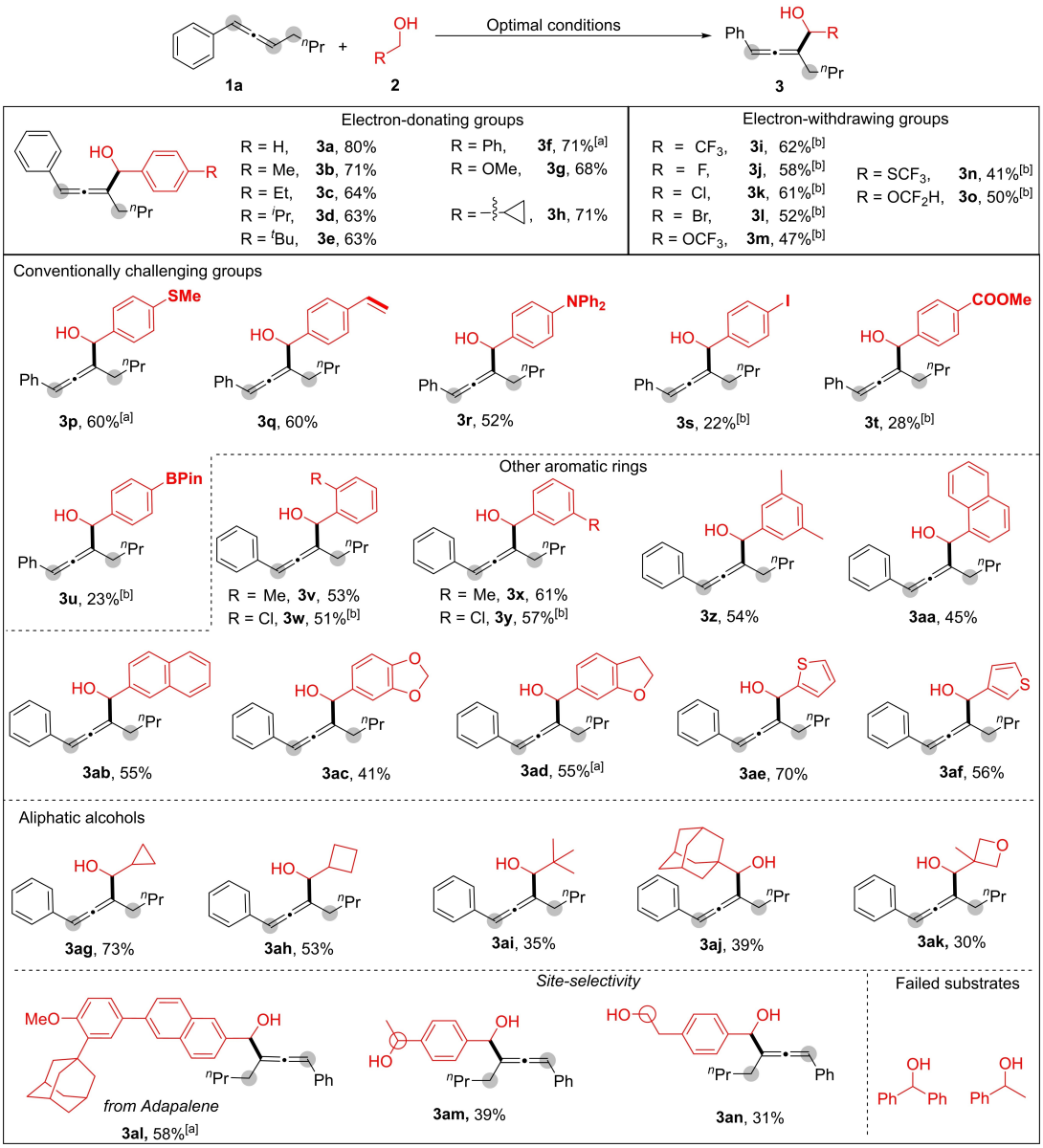
Applications in synthesis and mechanism investigation

The synthetic utility of our protocol was next explored with the direct derivatization of **3a** (Figure 3A). Interestingly, two types of dehydrated products **4** and **5** were selectively accessed. In the presence of trimethyl phosphate, a conjugated enyne product **4** was accessed in 66 %. On the contrary, using a catalytic Lewis acid $AuCl_3$ led to a cyclized product **5** via a Fridel-Crafts type reaction.

To shed light into the reaction mechanism, a cyclic voltammetry (CV) study, control experiments, and calculations were conducted. First, CV revealed that the anodic oxidation of benzyl alcohol was mediated by TEMPO with the detection of an obvious catalytic current (Figure S28). Second, we tested the electrochemical transformation in a

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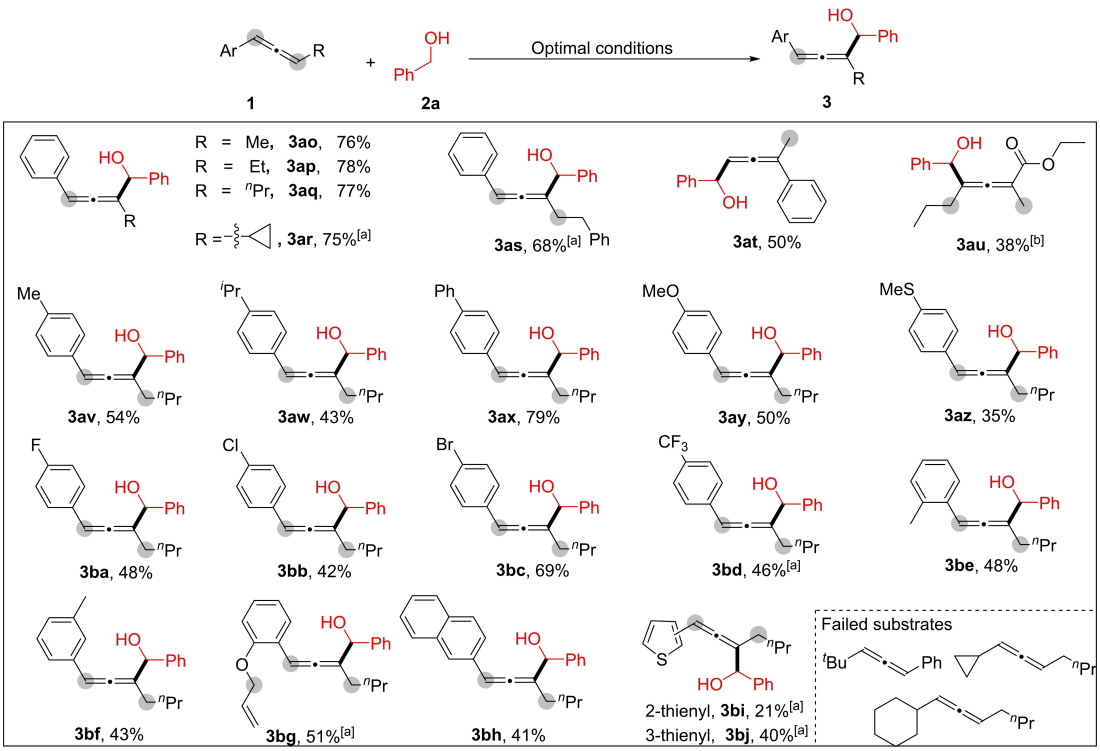


Scheme 1. Scope of alcohols. Reaction conditions: **1a** (1.0 mmol), **2** (0.5 mmol), **cat 1** (0.025 mmol), TEMPO (0.1 mmol), DMF (10 mL), $^{n}Bu_4NBF_4$ (1.0 mmol), graphite felt anode, nickel plate cathode, undivided cell, 0 °C, 20 mA, 4 h (5.97 F/mol), *dr* 1/1–15/1. [a] 1.5 mmol **1a** was used. [b] 1.5 mmol **1a** was used with **cat 2** as HER catalyst.

divided cell, and only 35 % yield of benzaldehyde was detected in the anodic chamber (Figure 3B, Eq. a). Further experiments replacing anodic catalyst TEMPO with excessive oxidant 2,2,6,6-tetramethyl-1-oxopiperidinium tetrafluoroborate only resulted in benzaldehyde (Eq. b), either. These results support that the electrochemical functionalization proceeds via a paired electrolysis pathway, and benzaldehyde should be a curial intermediate in the reaction. We verified this postulation by the experiment using benzaldehyde instead of benzyl alcohol (Eq. c). As expected, a moderate yield (60 %) of product **3a** was detected. The lower yield in the control experiments might be ascribed to the undesired Pinacol coupling of benzaldehyde, highlighting the efficiency of the current protocol. Third, control experiments and density functional theory (DFT) calculations were performed to rationalize the reaction selectivity enabled by catalytic HER (Figure 3C). Strong organic base ^{n}BuLi was employed to replace the electrochemical HER conditions (Eq. d). Fortunately, product **3a** and the related propargyl alcohol **3a'** were selectively observed, suggesting that the regioselectivity in the reaction should arise from the selective formation of carbanions. The regioisomer **3a** presumably forms via a Zimmermann–Traxler type transition state[57] and 1,3-metallotropic shift[58] process. This preference for the allenic C(sp^2)-H bonds was further interpreted with DFT calculations. The calculation results indicate that the pKa value of the C–H bond determines the reaction regioselec-

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Scheme 2. Scope of allenes. Reaction conditions: **1** (1.0 mmol), **2a** (0.5 mmol), **cat 1** (0.025 mmol), TEMPO (0.1 mmol), DMF (10 mL), $^{n}Bu_4NBF_4$ (1.0 mmol), graphite felt anode, nickel plate cathode, undivided cell, 0 °C, 20 mA, 4 h (5.97 F/mol), *dr* 1/1–3/1. [a] 1.5 mmol **1** was used. [b] 1.5 mmol **1** was used in the presence of 0.0125 mmol **cat 1**.

tivity (Table S3). The kinetic isotope effect (KIE) was also tested in the reaction by the intramolecular competing and parallel experiment (Eq. e–f).[59] It shows that the allenic C(sp^2)-H bond cleavage might be the rate-determining step in the reaction (Figure 3D). Finally, we conducted various control experiments to ascertain the real reactive catalyst in the reaction (Figure 3E). Using stoichiometric **cat 1** directly shut down the reaction in the absence of electricity (Eq. g). Replacing **cat 1** with in situ formed Co(II) species **A** affords the desired product **3a** in 64 % yield (Eq. h), whereas species **B** fails to give any product (Eq. i). These observations are consistent with the results of CV experiment (Figure 2E); the species **A** is identified as the real catalytic species, while **B** and precatalyst **cat 1** are inactive species in the reaction.

On the basis of the mechanism studies and related reports,[41–48] we proposed a plausible reaction mechanism (Figure 3F). Under cathodic reduction, [Co_2O_2] metallacycle **cat 1** is initially converted to cobalt(II) species **A** and **B**. As mentioned earlier, **A** is the real reactive species that could promote the dissociation of **1a** to afford carbanion **D** and a water-coordinated Co(II) complex **C**. Subsequently, a single electron transfer (SET) reduction of **C** generates a highly active Co(I) species **E**, which proceeds a subsequent proton relay to give **F**. Hydrogen evolution and regeneration of **A** can be accessed through the dimerization of hydrogen atom in the Co(III) complex **F**. On the other hand, benzyl alcohol was readily oxidized to benzaldehyde **G** with the mediation of TEMPO. The final product **3a** was generated via the nucleophilic addition between **G** and **D**.

Conclusion

In conclusion, a series of [Co_2O_2] metallacycle complexes were readily synthesized and well characterized. They were unveiled as robust hydrogen evolution catalysts, allowing a mild and highly selective reduction approach for allenic C(sp^2)-H bonds. Under paired electrolysis, a highly regioselective allene C(sp^2)-H functionalization with alcohols was unprecedentedly achieved. We rationalized the unconventional regioselectivity with various control experiments and DFT calculations. The further application of the [Co_2O_2] metallacycle catalysts has been actively studied in our laboratory.

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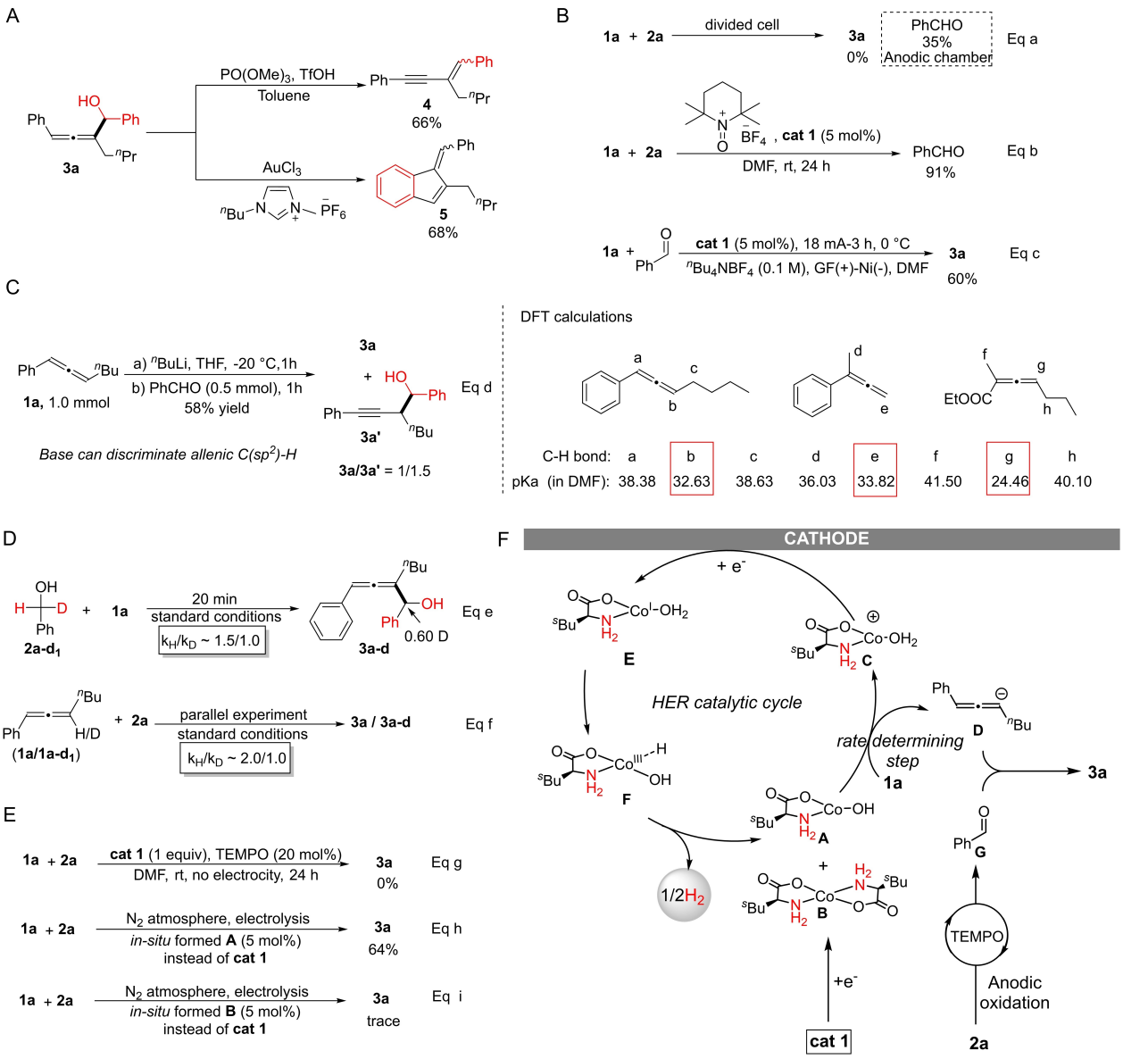


Figure 3. Derivatization and mechanism examination. (A) Derivatization of the product **3 a**. (B) Reaction intermediate and pathway investigation. (C) Investigation on the reaction selectivity. (D) Kinetic isotope effect study. (E) Identifying the reactive catalyst in the reaction. (F) Proposed reaction mechanism.

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Hydrogen evolution catalyst · $[Co_2O_2]$ metallacycle · Allene C(sp^2)-H · Electrochemical functionalization · Regioselectivity

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